

Question 1

1.

- a. False
- b. False
- c. True

2.

When the amount of clients is relatively small and the bandwidth can fit all the conversations, meaning there couldn't be a situation in which a client requires to establish a connection but there is not enough bandwidth to support it.

3.

Yes, each point of the network might have a different processing time and there could occur some congestion in one of the points of the network, meaning that there is a need for congestion control.

Question 2

For Network X (internal communication within Singapore), Danielle should acquire swift routers to optimize processing times within the short distance.

In this case, swift routers may be more beneficial. Swift routers are known for their shorter processing times, which can help minimize delay in routing data within a short distance.

For Network S (connection with the server farm in Sydney), Danielle should invest in high-speed communication cables to minimize propagation delay over the longer distance. For long-distance communication, the propagation speed of the communication medium becomes crucial.

Question 3

List of the time it takes for each part of the network:

time of queueing and processing = 2ms

$$\text{time of transmission at each point} = \frac{6 \cdot 8 \cdot 10^6}{1 \cdot 10^9} \cdot 10^3 = 48 \text{ms}$$

$$\text{time of propagation of } x \text{ meters} = \frac{x \cdot 10^3}{2 \cdot 10^8} \cdot 10^3 \text{ms}$$

Thus we can infer that the total time taken for the file to arrive from Y to BGU is 348 ms.

$$0 + 48 + 60 + 2 + 48 + 20 + 2 + 48 + 40 + 2 + 48 + 40 + 2 + 48 + 30 = 348 \text{ms}$$

Question 4

1. HTTP 1.1
2. persistent
3. Here are two possible responses:

```
HTTP/1.1 200 OK
Date: Sat, 21 Jan 2024 12:00:00 GMT
Server: Apache/2.4.49
Content-Type: text/html
Content-Length: 512
Connection: keep-alive

[Content of the requested /martignon/index.html]
```

```
HTTP/1.1 404 Not Found
Date: Sat, 21 Jan 2024 12:00:00 GMT
Server: Apache/2.4.49
Content-Type: text/html
Content-Length: 512
Connection: keep-alive

[Content of error page for 404 Not Found]
```

Question 5

1.

No.	Time	Source	Destination	Protocol	QUIC Pay	Length	Source Port	Destination Port	Info
13	1.883306000	10.100.102.35	10.100.102.1	DNS			107		Standard query 0x1534 HTTPS optimizationguide-p
15	1.883400000	10.100.102.35	10.100.102.1	DNS			107		Standard query 0xe6f4 A optimizationguide-pa.go
19	1.884273000	10.100.102.35	10.100.102.1	DNS			86		Standard query 0x4002 A www.reddit.com
23	1.884513000	10.100.102.35	10.100.102.1	DNS			86		Standard query 0x0f87 HTTPS www.reddit.com
32	1.896479000	10.100.102.1	10.100.102.35	DNS			205		Standard query response 0xe6f4 A optimizationgu
36	1.898424000	10.100.102.1	10.100.102.35	DNS			166		Standard query response 0x1534 HTTPS optimizati
42	1.900964000	10.100.102.1	10.100.102.35	DNS			139		Standard query response 0x4002 A www.reddit.com

Our IP is 10.100.102.35 and the IP of the local DNS server is 10.100.102.1 , we can infer that by knowing that the source of the DNS response is the local DNS server.

2.

```
> Frame 42: 139 bytes on wire (1112 bits), 139 bytes captured (1112 bits) on interfa
> Ethernet II, Src: SagemcomBroa_0d:09:97 (34:49:5b:0d:09:97), Dst: Intel_74:35:74 (
> Internet Protocol Version 4, Src: 10.100.102.1, Dst: 10.100.102.35
> Transmission Control Protocol, Src Port: 53, Dst Port: 50066, Seq: 1, Ack: 35, Len
> Domain Name System (response)
```

We can see that ethernet protocol is used for the link layer, IP is used for the network layer, TCP is used for transportation layer, and DNS is used for the application layer.

3.

Flags: 0x010 (ACK)

Window: 314

[Calculated window size: 160768]

[Window size scaling factor: 512]

Checksum: 0x15d8 [unverified]

The size of the receive window is 160768 bytes.

4.

- ▼ Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale,
 - ▼ TCP Option - Maximum segment size: 1460 bytes
 - Kind: Maximum Segment Size (2)
 - Length: 4
 - MSS Value: 1460
 - TCP Option - No-Operation (NOP)

Yes, the MSS has been configured for the connection, it is the maximum segment size. Once the MSS is established, it determines the maximum amount of payload data that can be included in each TCP segment without fragmentation.